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Electroacupuncture Reduces Duration of Postoperative Ileus After Laparoscopic Surgery for Colorectal Cancer

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BACKGROUND & AIMS: We investigated the efficacy of electroacupuncture in reducing the duration of postoperative ileus and hospital stay after laparoscopic surgery for colorectal cancer. **METHODS:** We performed a prospective study of 165 patients undergoing elective laparoscopic surgery for colonic and upper rectal cancer, enrolled from October 2008 to October 2010. Patients were assigned randomly to groups that received electroacupuncture (n = 55) or sham acupuncture (n = 55), once daily from postoperative days 1–4, or no acupuncture (n = 55). The acupoints Zusanli, Sanyinjiao, Hegu, and Zhigou were used. The primary outcome was time to defecation. Secondary outcomes included postoperative analgesic requirement, time to ambulation, and length of hospital stay. **RESULTS:** Patients who received electroacupuncture had a shorter time to defecation than patients who received no acupuncture (85.9 ± 36.1 vs 122.1 ± 53.5 h; $P < .001$) and length of hospital stay (6.5 ± 2.2 vs 8.5 ± 4.8 days; $P = .007$). Patients who received electroacupuncture also had a shorter time to defecation than patients who received sham acupuncture (85.9 ± 36.1 vs 107.5 ± 46.2 h; $P = .007$). Electroacupuncture was more effective than no or sham acupuncture in reducing postoperative analgesic requirement and time to ambulation. In multiple linear regression analysis, an absence of complications and electroacupuncture were associated with a shorter duration of postoperative ileus and hospital stay after the surgery. **CONCLUSIONS: In a clinical trial, electroacupuncture reduced the duration of postoperative ileus, time to ambulation, and postoperative analgesic requirement, compared with no or sham acupuncture, after laparoscopic surgery for colorectal cancer. ClinicalTrials.gov number, NCT00464425.**

Keywords: Colon Cancer; Therapy; Recovery Time; Gastrointestinal Motility.

Postoperative ileus is defined as a temporary delay of gastrointestinal motility after surgery that prevents effective transit of intestinal contents or tolerance of oral intake. It is associated commonly with colorectal surgery, and may adversely influence patients' recovery and prolong hospital stay.[1] Laparoscopic colorectal surgery has been shown by randomized trials and meta-analyses to be associated with better short-term clinical outcomes including faster return of gastrointestinal function than open surgery.[2–5] However, the duration of postoperative ileus (defined by the time to first bowel motion) is still reported to be as long as 4 days in the laparoscopic arm in most of these trials, which is just about 1 day earlier than that in the open arm.[2–4] Additional measures thus are necessary to further enhance the gastrointestinal recovery after laparoscopic colorectal surgery because this may have a positive impact on the duration of hospital stay as well as hospital costs. It is estimated that a reduction of the mean length of hospital stay by as little as 1 day may reduce the annual health care system costs in the United States by approximately $1 billion US dollars.[1]

Acupuncture is widely accepted in China as well as throughout the world as an effective treatment option for the management of postoperative nausea and vomiting and various functional gastrointestinal disorders.[6,7] Its role in treating postoperative ileus, however, is less clear, and data from the Chinese as well as the Western literature are scarce. In several animal studies using Sprague–Dawley rats, electroacupuncture (EA; a combined procedure with acupuncture and electric current stimulation) at acupoint Zusanli (stomach meridian ST-36) has been shown to accelerate colonic motility and promote colonic contractility via parasympathetic and cholinergic pathways.[8,9] There are only 2 small randomized studies identified from the Chinese literature addressing the role of acupuncture in treating postoperative ileus in human subjects. In a report of 26 patients undergoing abdominal surgery, 12 of 13 patients assigned to acupuncture at Zusanli showed recovery of normal bowel function within 72 hours after surgery, compared with only 6 of 13 patients in the control group.[10] In another randomized study of 39 patients, the time of first excretion after abdominal surgery in the acupuncture group (using acupoints Zusanli and Sanyinjiao) was significantly faster than that in the control group.[11] Unfortunately, the methodologic flaws associated with these studies including inappropriate and small sample size, unclear randomization method, and poorly defined outcomes have precluded the drawing of any strong and valid conclusions regarding the efficacy of acupuncture for postoperative ileus. Further

Abbreviations used in this paper: EA, electroacupuncture; LI, large intestine meridian; NA, no acupuncture; SA, sham acupuncture; SP, spleen meridian; ST, stomach meridian; TE, triple energizer meridian.

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good-quality clinical studies are necessary to confirm the therapeutic potential of EA in stimulating intestinal transit and resolving paralytic ileus after colorectal surgery.

We conducted a prospective randomized sham-controlled study that aimed to evaluate the efficacy of EA in reducing the duration of postoperative ileus and hospital stay after laparoscopic colorectal surgery.

Patients and Methods

Study Participants

We conducted the study from October 2008 through October 2010 at the Prince of Wales Hospital, a university teaching hospital in Hong Kong. The study protocol was approved by the Joint Chinese University of Hong Kong–New Territories East Cluster Clinical Research Ethics Committee, and registered with ClinicalTrials.gov, number NCT00464425. Inclusion criteria were as follows: consecutive patients undergoing elective laparoscopic resection of colonic and upper rectal cancer without the need for conversion, patients older than 18 years of age, patients with American Society of Anesthesiologists grading I–III, and patients who provided written informed consent. We excluded patients who underwent laparoscopic resection of midrectal and low rectal cancer, those who underwent total/proctocolectomy, those who underwent complex or combined laparoscopic procedures, those with stoma creation, those who developed intraoperative problems or complications, those who received epidural anesthesia or analgesia, those with a cardiac pacemaker, those who were allergic to acupuncture needles, and those who previously received acupuncture.

Study Design

Patients potentially eligible for the study were informed by the principal investigator about the study details the day before the scheduled surgery, and they followed a standard perioperative protocol, including preoperative mechanical bowel preparation. All laparoscopic surgeries were performed under general anesthesia. Our laparoscopic techniques have been described previously.[2] Patients were enrolled in the study if all the inclusion and exclusion criteria were satisfied after the laparoscopic surgery. Randomization was performed shortly after surgery by the co-investigator (W.W.L.), who is also an experienced physiotherapist-acupuncturist. Patients were randomized (using simple randomization) to receive either EA, sham acupuncture (SA), or no acupuncture (NA) in a 1:1:1 ratio. A sealed nonopaque envelope (produced according to the computer-generated random sequence) was opened to determine the limb of entry. The acupuncturist was the only individual who was aware of the treatment allocation; the patients randomized to the EA/SA groups and the outcome assessor were blinded to the treatment allocation (see later). It was not possible to blind the patients in the NA group. The patients randomized to the EA and SA groups underwent 1 session of acupuncture daily from postoperative day 1 until day 4, or until the time when the primary outcome had occurred, whichever was earlier. The postoperative management of all patients was standardized. Pethidine 1 mg/kg as postoperative analgesia was given every 4 hours on demand. Early ambulation was encouraged. Oral feeding was resumed as early as possible. Patients were discharged when they tolerated diet and were fully ambulatory.

Interventions

In the EA group, the acupoints relevant to the treatment of abdominal pain, abdominal distension, and constipation, including Zusanli (stomach meridian ST-36), Sanyinjiao (spleen meridian SP-6), Hegu (large intestine meridian LI-4), and Zhigou (triple energizer meridian TE-6), were identified before needle insertion (Figure 1). Selection of these acupoints was based on a consensus between the acupuncturist of the study and several professors of the Diploma Course of Clinical Acupuncture of the School of Professional and Continuing Education at the University of Hong Kong. Sterile acupuncture needles (length, 25 mm; diameter, 0.22 mm; Hwato; Suzhou Medical Appliance Factory, Suzhou, China) were inserted into these acupoints, with a depth of insertion of about 20 mm. The achievement of a radiating sensation with paresthesia, which is known as *de qi*, was indicative of effective needling. In the SA group, shorter needles (length, 13 mm; diameter, 0.22 mm; Hwato, Suzhou Medical Appliance Factory, Suzhou, China) were placed 15 mm away from the acupoints at a shallower depth of insertion (2 mm) to avoid *de qi*. This method has been used successfully as a valid sham control procedure in acupuncture trials for various medical problems.[12,13] No acupuncture was performed in the NA group. Electric stimulation was used in the EA group with the ES-160 6-channel programmable electroacupuncture device (Ito Company, Ltd, Tokyo, Japan). The frequency of the electric stimulation was set at 100 Hz, which has been shown by previous studies to be effective in enhancing the analgesic effect of acupuncture.[14] In the SA group, pseudostimulation was provided by deliberately connecting the needle to the incorrect output socket of the electroacupuncture device, and thus there was no flow of electric current. Patients could see the output light flashing but no current was transmitted throughout the procedure. Patients were told that the stimulation frequency selected was not perceivable by human beings. Each session of electroacupuncture lasted for 20 minutes.

Main Outcome Measures

The primary outcome of the study was the time to defecation, measured in hours, from the time the laparoscopic surgery ended until the first observed passage of stool. The secondary outcomes of the study included time of first passing flatus, time that the patients tolerated a solid diet, time to walk independently, duration of hospital stay, pain scores on visual analogue scale (from 0 which implied no pain at all, to 10 which implied the worst pain imaginable) on the first 3 postoperative days, and postoperative analgesic requirement. The data were recorded and measured by an independent research assistant. All authors had access to the study data and reviewed and approved the final manuscript.

Statistical Analysis

The data were analyzed according to the intention-to-treat principle. In this study, 2 null hypotheses were tested using the Student *t* test in a stepwise fashion. This methodology has been reported previously.[15] In the first step, it was investigated whether EA was more efficacious than NA in reducing the duration of postoperative ileus after laparoscopic colorectal surgery; and in a second step (only if the first null hypothesis was rejected), whether EA was more efficacious than SA. Secondary outcomes also were compared using the Student *t* test for parametric data and the Mann–Whitney *U* test for nonparametric data. Baseline categoric data were compared with the Pearson chi-squared test or the Fisher exact test. Multivariate analysis

A

Acupoint	Location	Indications	Depth
Zusanli (ST-36)	3 cun below the lower border of the patella, 1 finger breadth lateral to the anterior crest of the tibia, in the tibialis anterior muscle	Abdominal pain and distension Constipation Vomiting	0.5–1.2 cun
Sanyinjiao (SP-6)	3 cun directly above the tip of the medial malleolus, on the posterior border of the medial aspect of the tibia	Abdominal pain and distension	0.5–1 cun
Hegu (LI-4)	On the dorsum of the hand, between the 1st and 2nd metacarpal bones, approximately in the middle of the second metacarpal bone on the radial side	Main point for pain	0.5–1 cun
Zhigou (TE-6)	On the dorsum of the forearm, 3 cun proximal to the dorsal crease of the wrist, between the radius and ulna, on the radial side of the extensor digitorum muscle	Main point for constipation	0.8–1.2 cun

B

C D

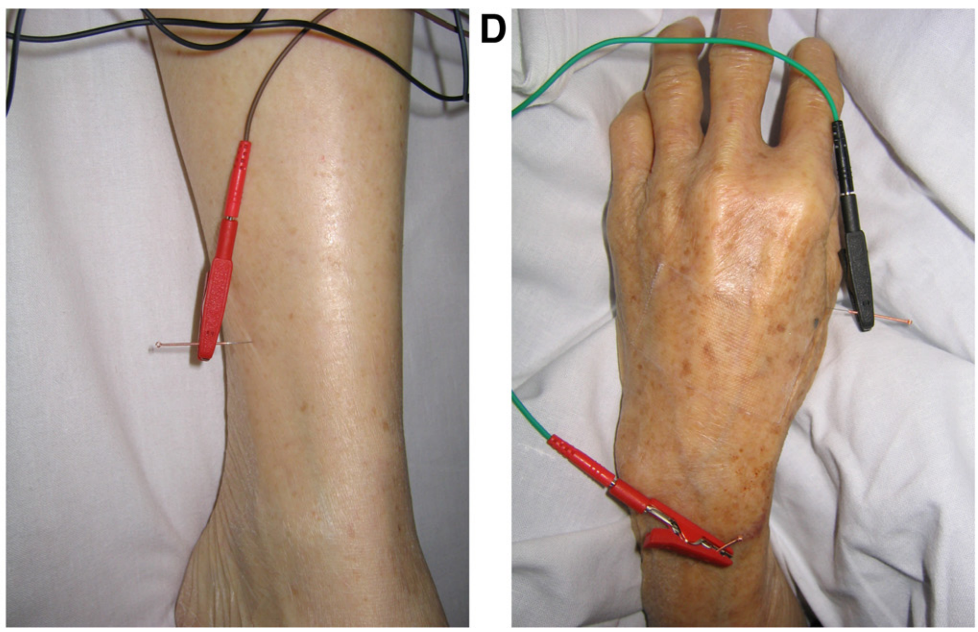

Figure 1. The acupoints used in this study. (*A*) Anatomic locations and indications of the acupoints. One cun is the distance between the 2 ends of the creases of the proximal and distal interphalangeal joints of the subject's index finger when flexed. (*B*) Zusanli, stomach meridian ST-36. (*C*) Sanyinjiao, spleen meridian SP-6. (*D*) Hegu, large intestine meridian LI-4, and Zhigou, triple energizer meridian TE-6.

with multiple linear regression was used to identify independent predictors of outcome measures. A *P* value less than .05 was considered statistically significant.

According to our previous randomized controlled trial on laparoscopic resection of rectosigmoid carcinoma, the mean time to first defecation in the laparoscopic arm was 4 days, with a standard deviation of 1.7 days.[2] With the assumption that the difference in mean time to first defecation between the EA and NA groups is 1 day or 24 hours, a sample size of 55 patients in each group was needed to yield a power of 80% with a significance level of .025 (2 pair-wise comparisons). Thus, a sample size of about 165 patients was needed for this study.

Results

Between October 2008 and October 2010, there were 208 consecutive patients who underwent elective laparoscopic resection of colonic and upper rectal cancer who were assessed for eligibility; of these, 43 patients were excluded after surgery. A total of 165 patients subsequently were enrolled and randomized to receive either EA (55 patients), SA (55 patients), or NA (55 patients) (Supplementary Figure 1). There was no withdrawal or dropout, and all recruited patients were available for analysis of primary and secondary outcomes. No adverse event related to the use of acupuncture was reported.

The study groups were similar with respect to demographic data and surgical details (Table 1). There was also no significant difference in the overall postoperative complication rates between the study groups (Table 2). Notably, more patients in the NA group developed prolonged paralytic ileus requiring nasogastric decompression and parenteral nutritional support when compared with the EA group, but the difference was not statistically signifi-

Table 1. Demographic Data and Surgical Details

	Electro acupuncture	Sham acupuncture	No acupuncture
Number of patients	55	55	55
Age, mean ± SD, *y*	67.4 ± 9.7	67.4 ± 10.7	68.5 ± 10.6
Sex, male/female	35/20	33/22	31/24
Body mass index, mean ± SD, *kg/m*^{2}	22.8 ± 2.9	22.9 ± 3.4	23.4 ± 3.1
ASA grading, I/II/III	17/36/2	14/30/11	11/32/12
Number of patients with comorbidities	29 (52.7%)	34 (61.8%)	37 (67.3%)
Types of surgery, RH/LH/SIG/AR	15/12/7/21	12/12/7/24	17/10/8/20
Surgical time, mean ± SD, *min*	157.3 ± 39.2	158.6 ± 46.5	164.1 ± 52.5
Median blood loss (range), *mL*	20 (0–200)	20 (0–200)	20 (0–600)

AR, anterior resection; ASA, American Society of Anesthesiologists; LH, left hemicolectomy; RH, right hemicolectomy; SD, standard deviation; SIG, sigmoid colectomy.

Table 3. Outcome Measures of Electroacupuncture Versus No Acupuncture Groups

	EA (n = 55)	NA (n = 55)	*P* value
Time to first passing flatus, *days*	2.0 ± 0.9	2.6 ± 1.1	.003
Time to first bowel motion, *h*	85.9 ± 36.1	122.1 ± 53.5	<.001
Time to resume normal diet, *days*	4.0 ± 1.1	4.8 ± 2.0	.010
Time to walk independently, *days*	2.8 ± 1.5	3.8 ± 1.8	.001
Hospital stay, *days*	6.5 ± 2.2	8.5 ± 4.8	.007
VAS pain score on day 1	5.6 ± 2.0	5.5 ± 2.3	.689
VAS pain score on day 2	3.2 ± 1.6	4.2 ± 2.0	.004
VAS pain score on day 3	2.1 ± 1.2	3.2 ± 1.8	<.001
Postoperative analgesic requirement, number of injections of 50-mg pethidine	2.7 ± 2.3	5.0 ± 4.5	.001

NOTE. All data are expressed as mean ± standard deviation and compared using the Student *t* test.
VAS, visual analogue scale.

cant (6 of 55 patients vs 3 of 55; $P = .489$). Similarly, no difference in rates of prolonged ileus as a complication was found between the EA and SA groups (3 of 55 patients vs 1 of 55; $P = .618$).

Electroacupuncture Versus No Acupuncture

Table 3 summarizes the outcome measures of the EA and NA groups. Compared with the NA group, the EA group had a significantly shorter time to defecation (85.9 ± 36.1 vs 122.1 ± 53.5 h; $P < .001$) and time to resume normal diet. The EA group also was significantly better than the NA group in all the other parameters measured, including shorter time to walk independently, lower pain scores, less analgesic requirement, and earlier hospital discharge. Therefore, the first null hypothesis of this study (ie, there was no difference in efficacy between EA and NA) was rejected successfully.

Electroacupuncture Versus Sham Acupuncture

To exclude the possibility of a placebo effect, the outcome measures of the EA group were compared further with those of the SA group, and the results are summarized in Table 4. The time to defecation of the EA group was found to be significantly shorter than that of the SA group (85.9 ± 36.1 vs 107.5 ± 46.2 h; $P = .007$), and hence the second null hypothesis of this study (ie, there was no difference in efficacy between EA and SA) also successfully was rejected. Other outcome measures including time to walk independently, pain scores, and analgesic requirement also were significantly better in the EA group. The duration of hospital stay, however, was similar between the EA and SA groups (6.5 ± 2.2 vs 6.8 ± 1.7 days; $P = .491$).

Table 2. Postoperative Complications

	EA (n = 55)	SA (n = 55)	NA (n = 55)	*P* value
Prolonged ileus	3	1	6	
Anastomotic bleeding	0	1	0	
Intra-abdominal collection	0	0	1	
Wound infection	2	0	2	
Chest infection	0	2	2	
Urinary tract infection	1	3	2	
Retention of urine	0	1	1	
Confusion	1	2	2	
Acute coronary syndrome	0	0	2	
Atrial fibrillation	0	1	1	
Cholangitis	1	0	0	
Total number of patients with complications	6 (10.9%)	5 (9.1%)	10 (18.2%)	.318[a]

NOTE. A patient could have more than 1 complication.
[a]Pearson chi-squared test.

Table 4. Outcome Measures of Electroacupuncture Versus Sham Acupuncture Groups

	EA (n = 55)	SA (n = 55)	*P* value
Time to first passing flatus, *days*	2.0 ± 0.9	2.3 ± 1.1	.095
Time to first bowel motion, *h*	85.9 ± 36.1	107.5 ± 46.2	.007
Time to resume normal diet, *days*	4.0 ± 1.1	4.1 ± 0.8	.695
Time to walk independently, *days*	2.8 ± 1.5	3.3 ± 1.1	.028
Hospital stay, *days*	6.5 ± 2.2	6.8 ± 1.7	.491
VAS pain score on day 1	5.6 ± 2.0	5.8 ± 1.9	.655
VAS pain score on day 2	3.2 ± 1.6	4.6 ± 2.0	<.001
VAS pain score on day 3	2.1 ± 1.2	3.4 ± 2.2	<.001
Postoperative analgesic requirement, number of injections of 50-mg pethidine	2.7 ± 2.3	5.2 ± 4.7	.001

NOTE. All data are expressed as mean ± standard deviation and compared using the Student *t* test.
VAS, visual analogue scale.

Table 5. Multiple Linear Regression Analysis

Variable	Coefficient B	Standard error	95% CI for B	*P* value
Independent predictors of time to defecation				
Postoperative complications	39.3	10.3	18.8–59.7	<.001
EA	−27.8	7.3	−42.3 to −13.4	<.001
Independent predictors of duration of hospital stay				
Postoperative complications	5.9	0.6	4.7–7.0	<.001
EA	−1.6	0.5	−2.5 to −0.6	.002
SA	−1.2	0.5	−2.2 to −0.2	.016

CI, confidence interval.

Independent Predictors of Outcome Measures

Because many confounding factors (eg, age, sex, body mass index, presence of medical comorbidities, types of surgery, surgical time, and postoperative complications) could have affected the gastrointestinal motility and duration of hospital stay, their effects were analyzed by multiple linear regression (Table 5). The presence of complications was an independent predictor of longer time to defecation (regression coefficient, 39.3; 95% confidence interval, 18.8–59.7; $P < .001$), whereas the use of EA predicted a shorter time to defecation (regression coefficient, −27.8; 95% confidence interval, −42.3 to −13.4; $P < .001$). Three factors were found to be independent predictors of shorter duration of hospital stay: absence of complications ($P < .001$), the use of EA ($P = .002$), and the use of SA ($P = .016$).

Discussion

Acupuncture has been used to treat gastrointestinal symptoms in China for thousands of years. It also increasingly is being accepted by clinicians and patients in the West as an effective treatment option for various functional gastrointestinal disorders. However, the role of acupuncture in treating postoperative ileus is less clear, and good-quality data from Chinese and Western literature are scarce.

To date, there has been only one properly conducted randomized controlled trial published in the English literature that examined the role of EA in preventing prolonged postoperative ileus after open surgery for colon cancer.[16] In this study by Meng et al,[16] 90 patients were randomized to receive EA or no acupuncture after surgery. The acupoints used were Zusanli, Zhigou, Shangjuxu (stomach meridian, ST-37), and Yanglingquan (gallbladder meridian, GB-34). This study failed to show any significant differences between the 2 groups in terms of time to first bowel movement and pain scores. The investigators acknowledged that one of the limitations of their study was the use of epidural anesthesia in the majority of patients, which might diminish the possible effects of acupuncture by blocking the afferent and efferent neural pathways.

In the present study, EA at acupoints including Zusanli, Sanyinjiao, Hegu, and Zhigou was more effective than NA and SA in stimulating early return of bowel function and reducing postoperative analgesic requirement after laparoscopic colorectal surgery. In addition, EA was found to be an independent predictor of shorter duration of postoperative ileus. We performed a large and rigorous randomized study that confirmed the efficacy of EA for postoperative ileus after laparoscopic colorectal surgery. Patients who had received epidural anesthesia or analgesia were excluded from our study, and, hence, in contrast to the previous study by Meng et al,[16] we were able to show the genuine beneficial effects of EA for postoperative ileus and pain. Furthermore, there is ample evidence to suggest that the use of opioid-based analgesia can stimulate peripheral opioid receptors within the gastrointestinal tract, which may exacerbate postoperative ileus after surgery.[17,18] Therefore, the significant reduction in postoperative pain and opioid-based analgesic consumption associated with the use of EA may be another factor that contributes to the reduction of duration of postoperative ileus in the EA group. However, it is difficult to ascertain whether the improved gastrointestinal function observed in the EA group is attributed primarily to a direct effect of EA on bowel motility or secondary to the fact that EA can reduce pain and enhance ambulation after surgery.

Another remarkable finding of this study was that EA also was found to be more effective than NA in reducing the duration of hospital stay, but there was no significant difference between EA and SA. On multivariate analysis, both EA and SA were independent predictors of shorter duration of hospitalization after laparoscopic colorectal surgery, with EA being a stronger predictor. It is indeed quite difficult to offer a plausible explanation for this observed therapeutic benefit of SA; the EA and SA groups had a similar hospital stay despite a shorter time to defecation and ambulation in the EA group. Admittedly, the sample size and power of this study may not be adequate to show significant differences in all the parameters measured between the EA and SA groups. Furthermore, the duration of hospital stay could have been affected by the underlying psychosocial background of the patients. For instance, during our routine clinical practice, we do commonly encounter patients who simply refuse to go home after surgery because of psychosocial reasons (eg, feeling unsafe to go home, no one to look after them at home, and so forth). Unfortunately, these parameters were not properly documented and hence were not evaluated in this study.

In designing clinical trials for postoperative ileus, it is important to define specific study end points that indicate when an ileus has resolved. A commonly used primary end point in these studies was the time to recovery of gastrointestinal function (gastrointestinal-3), a 3-component composite end point, as defined by the latter of the following 2 events: time until the patient first tolerates solid food and the time until the patient first passes either flatus or a bowel movement.[19] However, flatus often is regarded as an insensitive end point,[20] and the time to resume diet can be influenced by the patient's perception and easily manipulated by the attending clinician. We therefore adopted time to defecation as the single primary outcome measure of this study because it is more objective and can be recorded readily by the assessor without bias.

Selection of acupoints in the present study was based on expert consensus provided by qualified and experienced acupuncturists. The most commonly used acupoint in treating gastrointestinal disorders (including postoperative ileus) in clinical studies is Zusanli.[10,11,16] As indicated previously, EA at Zusanli has been shown to accelerate colonic transit and stimulate distal colonic motility via parasympathetic and cholinergic pathways in conscious rats.[8,9] Tabosa et al[21] showed that EA at Zusanli and Sanyinjiao significantly increased myoelectric activity of the small intestine in Wistar rats. Furthermore, preoperative EA at Zusanli with 100 Hz of electrical stimulation has been shown to reduce the postoperative analgesic requirement and associated side effects in patients undergoing lower abdominal surgery.[14] This analgesic effect induced by EA of high frequency (100 Hz) has been shown in animal studies to be mediated by κ-opioid receptors through the release of dynorphin.[22] The choice of the other acupoints used in this study also was well supported by published data from human studies: Zhigou for treating chronic functional constipation,[23] and Hegu and Sanyinjiao for alleviating abdominal pain.[24] We believe that the various therapeutic effects of EA at these acupoints have added together and contributed to the encouraging results of our study.

A recent development in elective colorectal surgery is the introduction of an enhanced recovery program after surgery, also referred to as the fast-track perioperative program. This multimodal program combines a number of perioperative elements aiming for a faster postoperative recovery and earlier discharge, and a reduction in surgical stress response.[25,26] The success of a fast-track program depends on a committed and experienced multidisciplinary team often comprising surgeons, anesthesiologists, physiotherapists, dietitians, and nurses; the implementation of this program can cost a large amount of money and resources.[27] On the other hand, only a single trained acupuncturist is required for our EA program. Compared with the complex elements of a fast-track program, EA is simpler to implement and less labor intensive, and yet it may still confer the same benefits to patients in terms of faster postoperative recovery and shorter hospital stay. Further studies providing head-to-head comparisons between the fast-tack program and the use of EA are imperative to confirm the validity of the earlier statements.

Our study had several limitations. First, the study population represented a highly selected group of patients who underwent uncomplicated elective laparoscopic resection of colonic and upper rectal cancer. Patients with midrectal and low rectal cancer and patients undergoing complex or combined laparoscopic procedures were excluded. These complicated cases were apparently more likely to develop prolonged ileus and morbidity after surgery, and it is also uncertain whether EA will be beneficial to them. Second, we did not use a fast-track perioperative program because it was not the standard care in our institution.[2] Although a fast-track perioperative program has been shown to enhance postoperative recovery after open colorectal surgery,[28] its role in laparoscopic colorectal surgery is still not fully established yet.[29] The possible combined effects of EA and the fast-track program on the clinical outcomes after laparoscopic colorectal surgery will be an important area for further research. Third, we did not conduct a cost-effectiveness analysis to evaluate the economic impact of the use of EA on the hospital system. Laparoscopic colorectal surgery is costly, and it is important to determine whether the faster postoperative recovery brought about by the use of EA will reduce the financial burden to the hospital/health care system and improve the cost effectiveness of the procedure. Further studies are needed to address this issue.

In conclusion, this randomized sham-controlled study suggests that electroacupuncture at acupoints including Zusanli, Sanyinjiao, Hegu, and Zhigou is more effective than no acupuncture and sham acupuncture in stimulating early return of bowel function and reducing postoperative analgesic requirements after laparoscopic colorectal surgery. Electroacupuncture is also more effective than no acupuncture in reducing the duration of hospital stay. The use of electroacupuncture is an independent predictor of shorter duration of postoperative ileus and hospital stay after laparoscopic colorectal surgery. Further studies are warranted to validate and generalize our findings, and to assess the impact of electroacupuncture on the cost-effectiveness of laparoscopic colorectal surgery.

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Gastroenterology* at www.gastrojournal.org, and at http://dx.doi.org/10.1053/j.gastro.2012.10.050.

References

1. Person B, Wexner SD. The management of postoperative ileus. Curr Probl Surg 2006;43:6–65.
2. Leung KL, Kwok SP, Lam SC, et al. Laparoscopic resection of rectosigmoid carcinoma: prospective randomised trial. Lancet 2004;363:1187–1192.
3. Guillou PJ, Quirke P, Thorpe H, et al. Short-term endpoints of conventional versus laparoscopic-assisted surgery in patients with colorectal cancer (MRC CLASICC trial): multicentre, randomised controlled trial. Lancet 2005;365:1718–1726.
4. Veldkamp R, Kuhry E, Hop WC, et al. Laparoscopic surgery versus open surgery for colon cancer: short-term outcomes of a randomised trial. Lancet Oncol 2005;6:477–484.
5. Reza MM, Blasco JA, Andradas E, et al. Systematic review of laparoscopic versus open surgery for colorectal cancer. Br J Surg 2006;93:921–928.

6. Ouyang H, Chen JD. Review article: therapeutic roles of acupuncture in functional gastrointestinal disorders. Aliment Pharmacol Ther 2004;20:831–841.
7. Takahashi T. Acupuncture for functional gastrointestinal disorders. J Gastroenterol 2006;41:408–417.
8. Iwa M, Matsushima M, Nakade Y, et al. Electroacupuncture at ST-36 accelerates colonic motility and transit in freely moving conscious rats. Am J Physiol Gastrointest Liver Physiol 2006;290: G285–G292.
9. Luo D, Liu S, Xie X, et al. Electroacupuncture at acupoint ST-36 promotes contractility of distal colon via a cholinergic pathway in conscious rats. Dig Dis Sci 2008;53:689–693.
10. Wan Q. Auricular-plaster therapy plus acupuncture at zusanli for postoperative recovery of intestinal obstruction. J Tradit Chin Med 2000;20:134–135.
11. Liu JX, Zhao Q. Effect of acupuncture on intestinal motion and sero-enzyme activity in perioperation. Zhong Xi Yi Jie He Za Zhi 1991;11:133–134, 156–157.
12. Lee SW, Liong ML, Yuen KH, et al. Validation of a sham acupuncture procedure in a randomised, controlled clinical trial of chronic pelvic pain treatment. Acupunct Med 2011;29:40–46.
13. Vincent C, Lewith G. Placebo controls for acupuncture studies. J R Soc Med 1995;88:199–202.
14. Lin JG, Lo MW, Wen YR, et al. The effect of high and low frequency electroacupuncture in pain after lower abdominal surgery. Pain 2002;99:509–514.
15. Brinkhaus B, Witt CM, Jena S, et al. Acupuncture in patients with chronic low back pain: a randomized controlled trial. Arch Intern Med 2006;166:450–457.
16. Meng ZQ, Garcia MK, Chiang JS, et al. Electro-acupuncture to prevent prolonged ileus: a randomized clinical trial. World J Gastroenterol 2010;16:104–111.
17. Kurz A, Sessler DI. Opioid-induced bowel dysfunction: pathophysiology and potential new therapies. Drugs 2003;63:649–671.
18. Holzer P. Opioids and opioid receptors in the enteric nervous system: from a problem in opioid analgesia to a possible new prokinetic therapy in humans. Neurosci Lett 2004;361:192–195.
19. Tan EK, Cornish J, Darzi AW, et al. Meta-analysis: alvimopan vs. placebo in the treatment of post-operative ileus. Aliment Pharmacol Ther 2007;25:47–57.
20. Bungard TJ, Kale-Pradhan PB. Prokinetic agents for the treatment of postoperative ileus in adults: a review of the literature. Pharmacotherapy 1999;19:416–423.
21. Tabosa A, Yamamura Y, Forno ER, et al. Effect of the acupoints ST-36 (Zusanli) and SP-6 (Sanyinjiao) on intestinal myoelectric activity of Wistar rats. Braz J Med Biol Res 2002;35:731–739.
22. Han JS. Acupuncture and endorphins. Neurosci Lett 2004;361: 258–261.
23. Zhang ZL, Ji XQ, Zhao SH, et al. Multi-central randomized controlled trials of electroacupuncture at Zhigou (TE 6) for treatment of constipation induced by stagnation or deficiency of qi. Zhongguo Zhen Jiu 2007;27:475–478.
24. Xu HY, Yang F, Zhu J, et al. Effect of electroacupuncture at Hegu (LI 4) and Sanyinjiao (SP 6) on short-term adverse effects of drug-induced abortion. Zhongguo Zhen Jiu 2007;27:103–105.
25. Kehlet H. Fast-track colonic surgery. Lancet 2008;371:791–793.
26. Kehlet H, Wilmore DW. Evidence-based surgical care and the evolution of fast-track surgery. Ann Surg 2008;248:189–198.
27. Gouvas N, Tan E, Windsor A, et al. Fast-track vs standard care in colorectal surgery: a meta-analysis update. Int J Colorectal Dis 2009;24:1119–1131.
28. Wind J, Polle SW, Fung Kon Jin PH, et al. Systematic review of enhanced recovery programmes in colonic surgery. Br J Surg 2006;93:800–809.
29. Vlug MS, Wind J, van der Zaag E, et al. Systematic review of laparoscopic versus open colonic surgery within an enhanced recovery programme. Colorectal Dis 2009;11:335–343.

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Conflicts of Interest

The authors disclose no conflicts.

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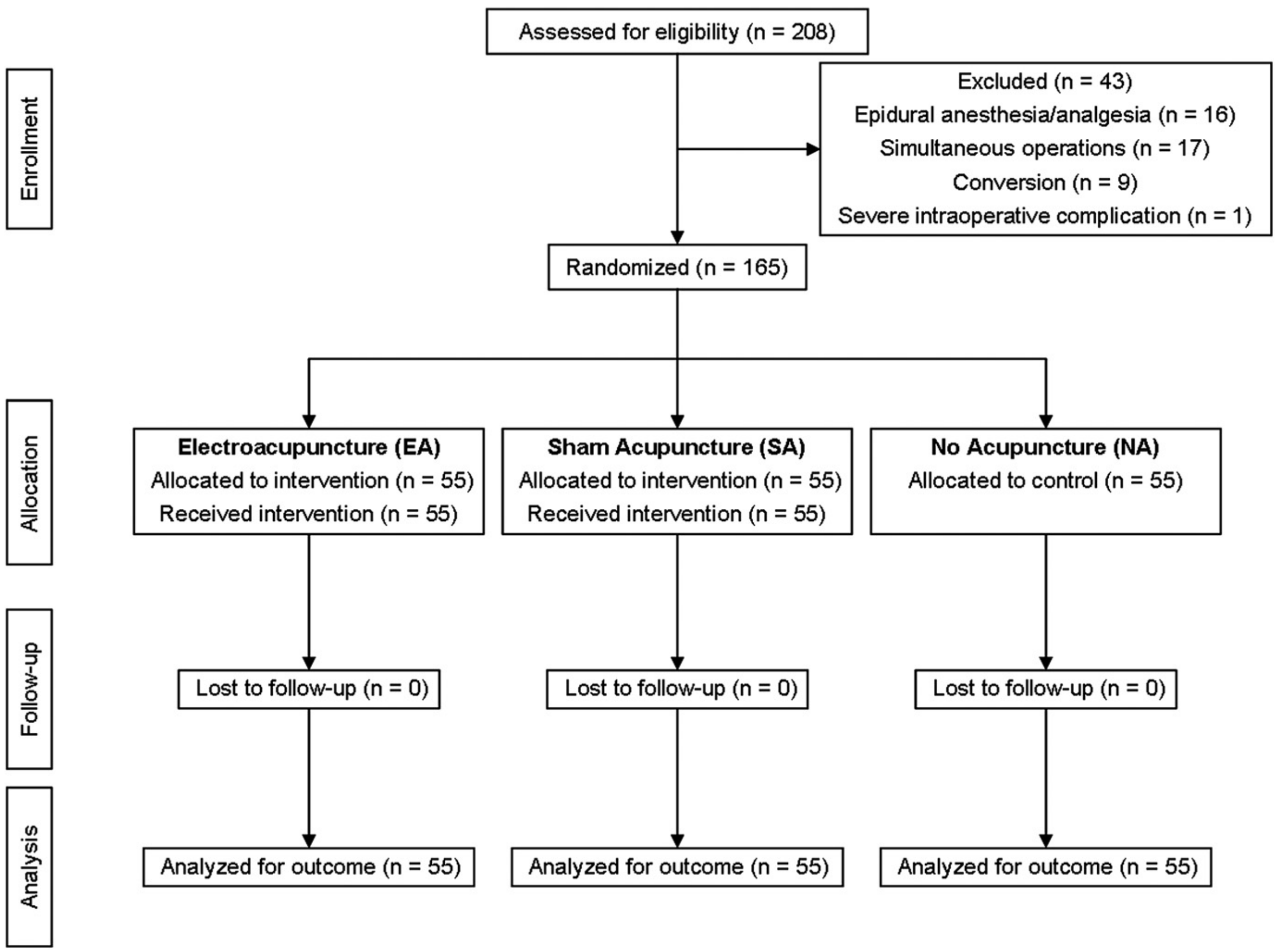

Supplementary Figure 1. Flow diagram.